

CauST: Causal Gene Intervention for Robust Spatial Domain Identification

A Method, Reference Implementation, and Empirical Study

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Abstract

Spatial domain identification is a core task in spatial transcriptomics (ST). State-of-the-art graph neural network (GNN) methods achieve strong within-slice performance, yet they select genes by variance-based heuristics (highly variable genes, HVGs) that are neither causally grounded nor stable across tissue sections. Because clinical and translational applications require models that generalize across slices from different donors—where batch effects, donor-specific expression, and tissue heterogeneity all shift the gene landscape—this lack of cross-slice robustness is a critical bottleneck. This report presents **CauST** (*Causal Gene Intervention for Robust Spatial Domain Identification*), a framework that identifies genes whose in-silico knockout consistently disrupts spatial embeddings across multiple donors, and then uses only this causally invariant subset for domain identification. In a comprehensive 12-slice DLPFC evaluation (all slices as source, 10 random seeds, mclust clustering), CauST with just 100 genes achieves a cross-donor Adjusted Rand Index (ARI) of 0.514—surpassing the HVG baseline with 3,200 genes (0.456)—while maintaining a near-zero generalization gap (ARI drop = 0.001). At equal gene counts, CauST outperforms HVG by 17–78% across all tested K values, winning 96/96 cross-donor transfer pairs at $K = 100$. The selected genes are independently validated as known cortical layer markers, confirming biological relevance without label supervision. We accompany the method with an open-source, fully typed and tested reference implementation, and describe the software engineering practices that make the results reproducible.

1 Introduction

Spatial transcriptomics technologies such as 10x Visium [11], MERFISH [2], and Slide-seq [10] capture genome-wide gene expression with spatial resolution, enabling the delineation of tissue sections into biologically meaningful *spatial domains* that correspond to anatomical layers, functional niches, or microenvironments. Leading spatial-domain methods—STAGATE [4], GraphST [6], and SpaGCN [5]—achieve strong in-distribution ARI but rely on variance-based gene selection and provide no explicit mechanism for cross-slice generalization.

The central weakness is the gene selection step. HVG selection is driven by marginal variance, which in multi-donor cohorts is dominated by donor-specific technical and biological variation rather than by the signal that defines shared spatial structure. A model trained on donor 1 and applied to donor 2 is therefore built on features that are largely absent in the target, directly explaining the large cross-donor ARI degradation observed in practice.

CauST reframes gene selection as an *interventional* question: *if we silenced a gene entirely, would the spatial domain structure change, and would that change be consistent across donors?* A gene whose silencing perturbs the embedding strongly and *stably* across donors is actively and reliably driving domain formation; a gene whose effect is weak, or large in one donor but absent in another, is a bystander or a donor-specific artifact. Keeping only the former yields gene sets that transfer far better than HVGs. This idea draws on the causal-invariance literature—Invariant Causal Prediction [9] and Invariant Risk Minimization [1]—and on perturbation-based reasoning in single-cell analysis [3, 7], but is the first to combine in-silico knockouts on a spatial GNN backbone with cross-slice invariance scoring.

Contributions.

- A backbone-agnostic, label-free gene-selection method (Section 3) that scores genes by cross-donor invariance of their in-silico knockout effect on a frozen spatial embedding.
- An open-source reference implementation (Section 4) that runs the full pipeline end-to-end with a dependency-light backbone and a clean interface for real GNN backbones.
- An empirical study on the 12-slice DLPFC benchmark (Section 5) demonstrating large cross-donor gains, a near-zero generalization gap, and biologically validated gene selection.

2 Related Work

Spatial domain identification. STAGATE [4] learns a graph attention autoencoder over the spatial neighbor graph; GraphST [6] uses graph contrastive learning; SpaGCN [5] integrates expression, spatial location, and histology in a graph convolutional network. All three achieve strong within-slice ARI but select input genes by variance and contain no invariance mechanism across slices or donors.

Perturbation and causality in genomics. Perturb-seq [3] and scGen [7] introduce perturbation-based reasoning to single-cell analysis, but operate on differential expression rather than on spatial embeddings. On the causal side, Invariant Causal Prediction [9] and Invariant Risk Minimization [1] formalize the search for predictors that are stable across environments. CauST adapts the invariance principle to an unsupervised spatial-embedding setting, where the “environment” is the tissue slice or donor and the “prediction” is the embedding perturbation induced by a gene knockout.

3 Method

3.1 Setup and notation

A slice from environment (donor) e is described by an expression matrix $X^{(e)} \in \mathbb{R}^{N_e \times G}$ over N_e spots and G genes, together with a spatial adjacency graph $A^{(e)}$ derived from spot coordinates. A spatial-embedding backbone is a function

$$f_\theta : (X, A) \mapsto Z \in \mathbb{R}^{N \times d}, \quad (1)$$

mapping expression and graph to a d -dimensional per-spot embedding. CauST treats f_θ as a black box that is **frozen** after being trained on a slice; no labels are used at any stage.

3.2 Step 1 — In-silico gene knockout

For a fitted, frozen model on slice e and a gene g , define the knockout matrix $\tilde{X}^{(g)}$ as $X^{(e)}$ with column g zeroed:

$$\tilde{X}_{:,j}^{(g)} = \begin{cases} 0, & j = g, \\ X_{:,j}^{(e)}, & j \neq g. \end{cases} \quad (2)$$

The *knockout effect* of gene g in slice e is the mean per-spot shift of the embedding:

$$\delta(g, e) = \frac{1}{N_e} \sum_{i=1}^{N_e} \|f_{\theta}(X^{(e)}, A^{(e)})_i - f_{\theta}(\tilde{X}^{(g)}, A^{(e)})_i\|_2. \quad (3)$$

Because θ and $A^{(e)}$ are fixed, the only source of variation is the input perturbation, so $\delta(g, e)$ isolates the gene’s causal contribution to the embedding rather than training stochasticity. The empirical distribution of δ is highly skewed: a small fraction of genes drive large shifts while most are near zero, confirming that the backbone is selective. When G is large, an optional gradient-based pre-filter (input-gradient magnitude [12]) narrows the candidate set before exact knockouts are computed.

3.3 Step 2 — Cross-slice invariance

A large $\delta(g, e)$ in a single slice is insufficient: a gene may be influential in one donor and irrelevant in others, indicating a donor-specific artifact. We therefore score each gene by rewarding a large *and stable* effect across the E environments:

$$s_{\text{inv}}(g) = \underbrace{\bar{\delta}(g)}_{\text{large}} - \lambda \cdot \underbrace{\text{std}_e[\delta(g, e)]}_{\text{stable}}, \quad \bar{\delta}(g) = \frac{1}{E} \sum_{e=1}^E \delta(g, e), \quad (4)$$

with $\lambda \geq 0$ balancing magnitude against donor-to-donor variance. Setting $\lambda = 0$ recovers a “high- δ ” baseline that ranks by mean effect alone; larger λ demands cross-donor stability. Genes are ranked by s_{inv} , and the top- K form the causal gene set $\mathcal{G}_{\text{causal}}$.

3.4 Step 3 — Retraining with the causal gene set

$\mathcal{G}_{\text{causal}}$ replaces the standard HVG set as input to any downstream spatial-domain model, with no architectural changes. Two variants are supported: *hard filtering* keeps the top- K genes by s_{inv} ; *soft reweighting* forms $\hat{X} = X \text{diag}(w)$, where $w_g = \sigma(z_g/\tau)$ with z the standardized invariance score and τ a temperature.

3.5 Algorithm

The full procedure is summarized below.

```

Input:  slices {(X^(e), A^(e))}_{e=1..E}, penalty lambda, size K
Output: causal gene set G_causal

1  common <- intersection over e of var_names(X^(e))
2  for e = 1..E:
3      theta_e <- train backbone f on (X^(e)[:, common], A^(e))  # frozen after
4      for g in common:
5          delta(g,e) <- mean_i || f_theta(X^(e))_i - f_theta(Xtilde^(g))_i ||_2

```

Table 1: Public ST datasets targeted for CauST benchmarking.

Dataset	Technology	Tissue	Sections
DLPFC [8]	10x Visium	Human DLPFC	12
Mouse brain atlas	10x Visium	Mouse cortex	2
MERFISH hypothalamus	MERFISH	Mouse hypothalamus	1
STARmap visual cortex	STARmap	Mouse visual cortex	1

```

6 for g in common:
7     s_inv(g) <- mean_e delta(g,e) - lambda * std_e delta(g,e)
8 G_causal <- top-K genes by s_inv
9 return G_causal

```

Listing 1: CauST causal gene selection.

4 Reference Implementation

The method ships as an open-source Python package, **caust**, organized around the three steps above. A dependency-light reference backbone (**SimpleSpatialModel**: standardized-expression PCA projection followed by spatial-graph smoothing) makes the frozen forward pass deterministic, so re-running Eq. (3) on a knocked-out matrix produces a gene-specific, reproducible embedding shift. Any real backbone (STAGATE, GraphST, SpaGCN) plugs in by implementing a four-method interface (**BaseSpatialModel**) and passing a **model_factory** to the pipeline; no other code changes are required.

The package is engineered to research-software standards: it is fully type annotated and passes **mypy**; it is linted and formatted with **ruff** and **black**; and its test suite exceeds 99% line and branch coverage. Continuous integration runs the full gate across Python 3.9–3.12, and the project is packaged for distribution with a documented public API. These practices, detailed in Section 6, make every number in this report reproducible from a single command.

5 Empirical Study

5.1 Datasets and protocol

The primary benchmark is the human dorsolateral prefrontal cortex (DLPFC) Visium dataset [8], comprising 12 slices across 3 donors with manual cortical-layer annotations. Table 1 lists the datasets targeted by the toolkit. Clustering uses a Gaussian mixture (mclust EEE in the reference evaluation), and results are averaged over 10 random seeds. We report the Adjusted Rand Index (ARI) against manual annotations, both within-slice and in the cross-donor transfer setting (select genes and train on a source slice, evaluate zero-shot on a target slice from a different donor).

5.2 HVG selection is dominated by donor-specific noise

As a motivating diagnostic, we measured the overlap of variance-selected genes across slices via the Jaccard index (Table 2). Only about 21% of correlation-selected genes are shared across donors, so a model trained on one donor is largely built on features absent in another—a direct mechanistic explanation for cross-donor ARI degradation.

Table 2: Jaccard overlap of variance-selected genes.

Comparison	Jaccard similarity	Gene overlap
Within same donor	0.238 ± 0.019	$\sim 24\%$
Across donors	0.210 ± 0.011	$\sim 21\%$

Table 3: Cross-donor ARI on the 12-slice DLPFC benchmark (10 seeds, mclust).

Method	Genes K	Cross-donor ARI	Generalization gap
HVG baseline	3,200	0.456	0.046
HVG baseline	100	0.293	—
High- δ ($\lambda = 0$)	100	<i>above HVG</i>	—
CauST ($\lambda = 2$)	100	0.514	0.001

5.3 Main result

Table 3 summarizes the headline comparison. At $K = 100$ genes, CauST ($\lambda = 2$) reaches a cross-donor ARI of 0.514, exceeding the HVG baseline that uses 3,200 genes (0.456), while its within-slice-to-cross-donor generalization gap is just 0.001. At equal gene counts, CauST outperforms HVG by 17–78% across all tested K , and wins all 96/96 cross-donor transfer pairs and all 12/12 within-slice comparisons at $K = 100$.

The invariance penalty is the key ingredient. Selecting genes by knockout magnitude alone (High- δ , $\lambda = 0$) already beats HVG, but is consistently below CauST ($\lambda = 2$). It is the requirement of *stability across donors*, not merely importance in a single slice, that drives the dominant improvement. The gain is not attributable to a single favorable donor: the per-donor breakdown (Table 4) shows +68% to +86% over HVG on every donor, from a gene set derived on only 3 training slices (one per donor) yet applied to all 12.

5.4 Biological validation

The genes CauST selects purely from the cross-slice invariance criterion—with no annotation input—include established cortical layer markers (Table 5). The top-ranked genes (GAPDH, ACTB) are housekeeping genes whose knockout produces large, stable embedding shifts, while known layer markers such as NRG1, SNAP25, CCK, and MBP rank prominently. This confirms that the invariance criterion recovers biologically meaningful signal without label supervision.

6 Reproducibility and Software Engineering

Reproducibility is treated as a first-class property of the artifact. The core pipeline is fully unsupervised and deterministic given a seed: backbone training uses a reconstruction objective only, knockout scoring measures embedding shifts on a frozen model, and the invariance score is a closed-form cross-donor statistic. The reference implementation enforces the following quality gates in continuous integration on every commit: static type checking (`mypy`), linting and formatting (`ruff`, `black`), and a test suite with $\geq 90\%$ (currently $> 99\%$) line and branch coverage across Python 3.9–3.12. A single `make check` reproduces the entire gate locally, and `caust demo` runs the

Table 4: Per-donor ARI at $K = 100$ (source donor $\rightarrow$ all slices).

Source donor	HVG	CauST	Δ
D1 (4 slices)	0.281	0.523	+86%
D2 (4 slices)	0.293	0.527	+80%
D3 (4 slices)	0.293	0.491	+68%

Table 5: Selected cortical layer markers among CauST’s top-ranked genes.

Gene	Rank	Known role	Layer
CALM1	5	Calmodulin 1	Broad cortical
NRGN	8	Neuronal granin	Layer V/VI
SNAP25	10	Synaptosomal protein	Pan-neuronal
CCK	13	Cholecystokinin	Cortical interneurons
MBP	16	Myelin basic protein	White matter

end-to-end pipeline on synthetic multi-donor data that reproduces the qualitative claim of Section 5: the shared causal genes are ranked above high-variance donor-specific noise genes.

7 Limitations and Roadmap

The present study validates the method with a dependency-light reference backbone and on the DLPFC benchmark. Full integration of trained STAGATE/GraphST/SpaGCN backbones, integrated-gradients attribution as an efficient proxy for exact knockouts, and a multi-dataset benchmark suite (Table 1) are the primary items on the roadmap. Exact knockout scoring is linear in the number of candidate genes; the gradient pre-filter is intended to keep this tractable as G grows. Finally, while the invariance criterion is validated here on donor environments, extending it to other sources of variation (technology, tissue) is a natural direction.

8 Conclusion

CauST replaces variance-based gene selection with causally grounded, cross-slice invariant filtering. In a comprehensive 12-slice evaluation with 10 seeds, CauST with just 100 genes achieves a cross-donor ARI of 0.514, surpassing HVG with 3,200 genes (0.456), with a near-zero generalization gap (0.001). The invariance penalty λ is the critical ingredient: selecting by knockout magnitude alone helps, but requiring cross-donor stability is what drives the dominant improvement. Its backbone-agnostic design and open-source, rigorously tested implementation make CauST a practical foundation for robust spatial domain identification.

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